CentraleSupélec

Data Analytics for Risk and Reliability

**Identifying Component Reliability Impacts on System Reliability from Observational Data**

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## TASK ONE: IMPACT OF DESIGN CHANGES ON RELIABILITY

We used the processed repair history dataset to compare reliability at 500 hours after a repair across five product models V1E1, V1E2, V1E3, V2E2, and V2E3. Each record contains time since previous repair, a right censoring indicator, and other baseline information such as product type, repair center, and age after repair. Our main empirical target is the survival probability R(500) = P(T > 500), where T is the time to the next failure or censoring. To estimate R(500) for each model, we used the Kaplan Meier method with time since previous repair as the duration and event observed = 1 minus censored as the event indicator. We extracted the survival probability at 500 hours for each model and used a nonparametric bootstrap with 300 resamples within each model to obtain 95 percent confidence intervals.

| | model | n | R500 | CI_low | CI_high |
|---|---|---|---|---|---|
| 3 | V2E2 | 3646 | 0.971941 | 0.967103 | 0.976409 |
| 4 | V2E3 | 2468 | 0.953576 | 0.944918 | 0.961865 |
| 2 | V1E3 | 725 | 0.934753 | 0.917074 | 0.953989 |
| 0 | V1E1 | 2056 | 0.921687 | 0.909244 | 0.932513 |
| 1 | V1E2 | 3925 | 0.873552 | 0.862066 | 0.884009 |

Table-1: Empirical reliability at 500 hours by model.

The empirical results show clear differences. The estimated R_hat(500) values are V2E2 = 0.9719 with 95 percent CI [0.9671, 0.9764], V2E3 = 0.9536 with [0.9449, 0.9619], V1E3 = 0.9348 with [0.9171, 0.9540], V1E1 = 0.9217 with [0.9092, 0.9325], and V1E2 = 0.8736 with [0.8621, 0.8840]. Based on these numbers, the empirical ranking from best to worst is V2E2, then V2E3, then V1E3, then V1E1, and finally V1E2. To show the differences over time and not only at 500 hours, we can also plot the Kaplan Meier survival curves for all models. Even though Kaplan Meier accounts for right censoring, this is still observational data, so differences between models might reflect differences in which units end up in each model group, not only the design itself. To check how comparable the model groups are, we made a summary by model including sample size, censoring rate, typical time since previous repair, age after repair, and the number of repair centers represented.

| model | n | censor_rate | median_time | mean_time | mean_age | median_age | n_centers |
|---|---|---|---|---|---|---|---|
| V1E2 | 3925 | 0.466497 | 1199.0 | 1277.311083 | 2255.899618 | 2439.0 | 6 |
| V2E2 | 3646 | 0.679923 | 1430.0 | 1444.087767 | 1560.882611 | 1524.0 | 5 |
| V2E3 | 2468 | 0.939627 | 652.0 | 636.185575 | 1140.557131 | 944.0 | 4 |
| V1E1 | 2056 | 0.242704 | 1246.5 | 1585.221790 | 1918.486868 | 1615.5 | 5 |
| V1E3 | 725 | 0.937931 | 504.0 | 494.532414 | 3189.619310 | 3072.0 | 2 |

Table-2: Model group comparability diagnostics.

The comparability table suggests important imbalances. Censoring rates are very different between models, for example V2E3 and V1E3 are heavily censored, while V1E1 has much lower censoring. Age after repair also differs a lot across models, with V1E3 tending to have older units than V2E3. Repair center coverage is uneven as well, with some models appearing in fewer centers than others. These differences can bias a simple comparison because failure risk may depend on age, center, and other baseline characteristics, not only on the model. To move from association to an adjusted estimate, we defined a binary outcome Y that equals 1 if a failure happens before 500 hours. For units censored before 500 hours, Y is not observed, so we handled censoring using inverse probability of censoring weighting. We estimated the censoring survival function G(u) = P(C > u) and used weights w = 1 divided by G(min(T, 500)). With these weights, we fit a weighted logistic regression for P(Y = 1 given model and W), where W included product type, repair center, age after repair, and repair operation number. Then we used standardization by setting model = m for everyone, predicting failure risk for each row, averaging, and converting back to reliability using R do(m) at 500 equals 1 minus the average predicted risk.

| | |
|---|---|
| V2E3 | 0.962121 |
| V1E1 | 0.950123 |
| V2E2 | 0.943638 |
| V1E2 | 0.873360 |
| V1E3 | 0.637538 |

Table-3: Adjusted reliability at 500 hours by model using IPCW weighted standardization.

The adjusted results give R_do(500) estimates of V2E3 = 0.9621, V1E1 = 0.9501, V2E2 = 0.9436, V1E2 = 0.8734, and V1E3 = 0.6375. The ranking changes compared to the empirical Kaplan Meier ranking, and V1E3 becomes much lower after adjustment. This big change is consistent with the earlier imbalance checks showing that V1E3 has a different case mix, especially higher age after repair, limited center coverage, and heavy censoring. Because adjusted estimates can become unstable when overlap is weak, it is also reasonable to report a short weight diagnostic such as high percentiles of the IPCW weights to show whether one model is driving extreme weights.

Overall, the empirical analysis suggests V2E2 has the highest observed R(500) and V1E2 the lowest, but the comparability diagnostics show that the model groups differ in censoring, age after repair, and repair center mix, so adjustment is needed. After applying IPCW weighting and standardization, the adjusted results favor V2E3 at 500 hours, while V1E3 appears substantially worse after controlling for observed differences and censoring.

## TASK TWO: IMPACT OF REPAIR CENTER ON RELIABILITY

We did Task 2 to check whether the repair center affects reliability at 500 hours after a repair. The dataset includes repair center, time since previous repair, and a censoring indicator, plus baseline information like model, product type, age after repair, and repair operation number. The main idea is to compare centers in two ways. First, we compare what we observe directly in the data. Then we do an adjusted analysis because centers may receive different kinds of units, which can make the raw comparison unfair.

For the empirical comparison, we treated time since previous repair as the survival time and used event observed equals 1 minus censored, so failures are events and censored observations are right censored. For each repair center, we estimated reliability at 500 hours using the Kaplan Meier method, meaning we took the survival probability at 500.

| | repair_center | n | R500_emp_KM |
|---|---|---|---|
| 4 | center_4 | 20 | 1.000000 |
| 5 | center_5 | 2 | 1.000000 |
| 6 | center_6 | 1 | 1.000000 |
| 0 | center_0 | 7331 | 0.947078 |
| 3 | center_3 | 93 | 0.946237 |
| 2 | center_2 | 4829 | 0.899261 |
| 1 | center_1 | 544 | 0.883118 |

Table-4: Empirical reliability at 500 hours by repair center (Kaplan Meier)

We report the Kaplan Meier estimate of R(500) for each repair center. We do not include confidence intervals here because several centers have very small sample sizes, so uncertainty is better discussed using the case mix and follow up diagnostics in Table 5. We should be careful interpreting centers with very small n or with 100 percent censoring, because an estimate near 1.00 can occur simply because no failures were observed during follow up, not because the center is truly better.

Next, we checked whether centers are comparable, because this is not randomized data. Different centers might handle different types of products or different situations, and that alone could change failure risk. To show this, we made a center level summary that includes sample size, censoring rate, average and median time since previous repair, and average and median age after repair.

| repair center | n | censor_rate | median_time | mean_time | mean_age | median_age | model_nunique | product_type_nunique |
|---|---|---|---|---|---|---|---|---|
| **center_0** | 7331 | 0.586687 | 1272.0 | 1383.845587 | 1396.794025 | 1279.0 | 4 | 2 |
| **center_2** | 4829 | 0.636778 | 821.0 | 915.402361 | 2423.024850 | 2378.0 | 5 | 2 |
| **center_1** | 544 | 0.608456 | 1192.5 | 1256.996324 | 2525.827206 | 2485.0 | 5 | 2 |
| **center_3** | 93 | 0.838710 | 2382.0 | 2239.440860 | 2710.354839 | 2611.0 | 3 | 2 |
| **center_4** | 20 | 1.000000 | 80.0 | 281.300000 | 2075.050000 | 1785.5 | 3 | 2 |
| **center_5** | 2 | 1.000000 | 1003.0 | 1003.000000 | 2517.500000 | 2517.5 | 1 | 1 |
| **center_6** | 1 | 1.000000 | 1292.0 | 1292.000000 | 3829.000000 | 3829.0 | 1 | 1 |

Table-5: Repair center case mix and follow up diagnostics

We also looked at how many different models and product types of each center handles. If Table 5 shows big differences in age, censoring, or model mix across centers, then a simple Kaplan Meier comparison might mostly reflect case mix rather than repair quality.

Finally, we did an adjusted analysis to estimate what reliability would look like if we could assign repairs to a specific center while keeping the same overall population. We defined the outcome Y as 1 if a failure happens before 500 hours and 0 otherwise. Y is unknown if an item is censored before 500 hours, so we handled censoring using inverse probability of censoring weights. We estimated the censoring survival function and used weights equal to 1 divided by that survival probability at min(time, 500). Then we fit a weighted logistic regression for the probability of failing before 500 as a function of repair center and baseline covariates, using model, product type, age after repair, and repair operation number as adjustment variables.

To get the adjusted reliability for each center, we set the repair center to that center for all rows, predicted the failure probability, averaged those predictions, and converted back to reliability using 1 minus the average failure probability.

```
center_0    0.961325
center_4    0.902859
center_5    0.883040
center_6    0.882437
center_1    0.827474
center_3    0.802956
center_2    0.776752
```

Table-6: Adjusted reliability at 500 hours by repair center.

In short, Table 4 shows the raw center differences at 500 hours, Table 5 shows whether those differences could be explained by different case mix and follow up, and Table 6 shows the adjusted comparison after controlling for measured variables and censoring. If a center's ranking changes a lot from Table 4 to Table 6, that suggests the raw differences were driven more by case mix than by the center itself.